

Evaluating Algebraic Expressions (Part I)

Name
Date
Course

→
→
PEMDAS

① $2x-3, x=4$ ② $5-6y, y=2$ ③ $-3z+1, z=10$

$$2(4)-3$$

$$5-6(2)$$

$$-3(10)+1$$

$$8-3$$

$$5-12$$

$$-30+1$$

$$\boxed{5}$$

$$\boxed{-7}$$

$$\boxed{-29}$$

④ $-3+9x, x=1$ ⑤ $5y+7, y=2$ ⑥ $2-4z, z=3$

$$-3+9(1)$$

$$5(2)+7$$

$$2-4(3)$$

$$-3+9$$

$$10+7$$

$$2-12$$

$$\boxed{6}$$

$$\boxed{17}$$

$$\boxed{-10}$$

⑦ $-2x-3, x=4$ * ⑧ $-y+6, y=2$ ⑨ $3-z, z=11$

$$-2(4)-3$$

$$-1y+6$$

$$3-1z$$

$$-8-3$$

$$-1(2)+6$$

$$3-1(11)$$

$$\boxed{-11}$$

$$-2+6=\boxed{4}$$

$$3-11=\boxed{-8}$$

⑩ $-x+9, x=7$ ⑪ $8-y, y=13$ ⑫ $-z+3, z=5$

$$-1x+9$$

$$8-1y$$

$$-1z+3$$

$$-1(7)+9$$

$$8-1(13)$$

$$-1(5)+3$$

$$-7+9$$

$$8-13$$

$$-5+3$$

$$\boxed{2}$$

$$\boxed{-5}$$

$$\boxed{-2}$$

⑬ $-x-5, x=7$ ⑭ $-2-y, y=6$ ⑮ $-z-1, z=1$

$$\begin{aligned} -1x-5 \\ -1(7)-5 \\ -7-5 \\ \boxed{-12} \end{aligned}$$

$$\begin{aligned} -2-1y \\ -2-1(6) \\ -2-6=\boxed{-8} \end{aligned}$$

$$\begin{aligned} -1z-1 \\ -1(1)-1 \\ -1-1=\boxed{-2} \end{aligned}$$

⑯ $-10-x, x=8$ * ⑰ $4x+2, x=-3$ ⑱ $-4z-5, z=-2$

$$-10-1x$$

$$4(-3)+2$$

$$-4(-2)-5$$

$$\begin{aligned} -10-1(8) \\ -10-8 \\ \boxed{-18} \end{aligned}$$

$$-12+2$$

$$\boxed{-10}$$

$$8-5$$

$$\boxed{3}$$

⑲ $-5z+7, z=-1$ ⑳ $-1+2x, x=-2$ ㉑ $-6y-3, y=-6$

$$-5(-1)+7$$

$$-1+2(-2)$$

$$-6(-6)-3$$

$$5+7$$

$$-1-4$$

$$36-3$$

$$\boxed{12}$$

$$\boxed{-5}$$

$$\boxed{33}$$

㉒ $8z-2, z=-2$ * ㉓ $3-2y, y=-4$ ㉔ $-2-9x, x=-2$

$$8(-2)-2$$

$$3-2(-4)$$

$$-2-9(-2)$$

$$-16-2$$

$$3-(-8)$$

$$-2-(-18)$$

$$\boxed{-18}$$

$$3+8$$

$$-2+18$$

$$\boxed{11}$$

$$\boxed{16}$$

$$(25) 1-4z, z=-10 \quad (26) -10-3y, y=-3 \quad (27) -x+1, x=-3$$

$$1-4(-10)$$

$$-10-3(-3)$$

$$-1x+1$$

$$1+40$$

$$-10+9$$

$$-1(-3)+1$$

$$\boxed{41}$$

$$\boxed{-1}$$

$$\begin{array}{c} 3+1 \\ \boxed{4} \end{array}$$

$$(28) -z-4, z=-1 \quad (29) 7-y, y=-2 \quad (30) -3-x, x=-8$$

$$-1z-4$$

$$7-1y$$

$$-3-1x$$

$$-1(-1)-4$$

$$7-1(-2)$$

$$-3-1(-8)$$

$$1-4$$

$$7+2$$

$$-3+8$$

$$\boxed{-3}$$

$$\boxed{9}$$

$$\boxed{5}$$

$$(31) -z+3, z=-5 \quad (32) 11-x, x=-8 \quad (33) -y-5, y=-2$$

$$-1z+3$$

$$11-1x$$

$$-1y-5$$

$$-1(-5)+3$$

$$11-1(-8)$$

$$-1(-2)-5$$

$$5+3$$

$$11+8$$

$$2-5$$

$$\boxed{8}$$

$$\boxed{19}$$

$$\boxed{-3}$$

$$(34) -9-z, z=-1 \quad (35) \frac{z}{2}+1, z=18 \quad (36) 5+\frac{x}{3}, x=-1$$

$$-9-1z$$

$$\text{PEMDAS} \quad \frac{18}{2}+1$$

$$5+(\frac{-1}{3})$$

$$-9-1(-1)$$

$$9+1$$

$$\frac{5}{3}-\frac{1}{3}$$

$$-9+1$$

$$\boxed{10}$$

$$\frac{15}{3}-\frac{1}{3}$$

$$\boxed{\frac{14}{3}}$$

$$(37) \frac{y}{4}-3, y=-12 \quad (38) -\frac{3}{z}+2, z=5 \quad (39) 8+\frac{x}{5}, x=-25$$

$$\frac{-12}{4}-3$$

$$-\frac{3}{5}+\frac{2}{1}$$

$$8+(\frac{-25}{5})$$

$$-3-3$$

$$-\frac{3}{5}+\frac{10}{5}$$

$$8+(-5)$$

$$\boxed{-6}$$

$$\boxed{\frac{7}{5}}$$

$$\boxed{3}$$

$$(40) -1-\frac{y}{8}, y=7 \quad (41) \frac{x}{7}+4, x=-56 \quad (42) \frac{6}{z}-2, z=-11$$

$$-1-\frac{7}{8}$$

$$\frac{-56}{7}+4$$

$$-\frac{6}{11}-\frac{2}{1}$$

$$-\frac{8}{8}-\frac{7}{8}$$

$$-8+4$$

$$-\frac{6}{11}-\frac{22}{11}$$

$$\boxed{-\frac{15}{8}}$$

$$\boxed{-4}$$

$$\boxed{-\frac{28}{11}}$$

$$(43) -\frac{x}{3}+2, x=-2 \quad (44) 5-\frac{y}{2}, y=-4 \quad (45) -\frac{z}{2}+6, z=-3$$

$$-\frac{(-2)}{3}+2$$

$$5-\frac{(-4)}{2}$$

$$-\frac{(-3)}{2}+6$$

$$\frac{2}{3}+\frac{2}{1}$$

$$5-(-2)$$

$$\frac{3}{2}+\frac{6}{1}$$

$$\frac{2}{3}+\frac{6}{3}=\boxed{\frac{8}{3}}$$

$$5+2$$

$$\frac{3}{2}+\frac{12}{2}$$

$$\boxed{7}$$

$$\boxed{\frac{15}{2}}$$

$$(46) -3-\frac{x}{6}, x=-5 \quad (47) -\frac{y}{2}+1, y=-2 \quad (48) 4-\frac{z}{2}, z=-7$$

$$-3-\frac{(-5)}{6}$$

$$-\frac{(-2)}{2}+1$$

$$4-\frac{(-7)}{2}$$

$$-\frac{3}{1}+\frac{5}{6}$$

$$-(-1)+1$$

$$4+\frac{7}{2}$$

$$-\frac{18}{6}+\frac{5}{6}$$

$$1+1$$

$$\frac{8}{2}+\frac{7}{2}$$

$$\boxed{-\frac{13}{6}}$$

$$\boxed{2}$$

$$\boxed{\frac{15}{2}}$$

$$\begin{array}{lll} \textcircled{49} \frac{x+2}{3} - 1, x=1 & \textcircled{50} 4 + \frac{7-y}{2}, y=10 & \textcircled{51} \frac{-2x-5}{7}, x=-1 \\ \frac{1+2}{3} - 1 & 4 + \frac{7-10}{2} & \frac{-2(-1)-5}{7} \\ \frac{3}{3} - 1 & 4 + \left(-\frac{3}{2}\right) & \frac{2-5}{7} \\ 1-1 & 4 - \frac{3}{2} & \boxed{-\frac{3}{7}} \\ \boxed{0} & \boxed{\frac{5}{2}} & \end{array}$$

$$\begin{array}{lll} \textcircled{52} \frac{2z-1}{5}, z=-3 & \textcircled{53} \frac{3}{x-1}, x=-2, & \textcircled{54} \frac{3y-1}{2-4y}, y=2 \\ \frac{2(-3)-1}{5} & \frac{3}{-2-1} & \frac{3(2)-1}{2-4(2)} \\ \frac{-6-1}{5} & \frac{3}{-3} & \frac{6-1}{2-8} \\ \boxed{-\frac{7}{5}} & \boxed{-1} & \boxed{\frac{5}{-6}} \end{array}$$

$$\begin{array}{lll} \textcircled{55} 5(x+2)^2 + 3, x=1 & \textcircled{56} 6 - 2(y-4)^2, y=2 & \textcircled{57} 1 + 2(1-z)^2, z=-1 \\ 5(1+2)^2 + 3 & 6 - 2(2-4)^2 & 1 + 2(1-(-1))^2 \\ 5(3)^2 + 3 & 6 - 2(-2)^2 & 1 + 2(1+1)^2 \\ 5(9) + 3 & 6 - 2(4) & 1 + 2(2)^2 \\ 45 + 3 & 6 - 8 & 1 + 2(4) \\ \boxed{48} & \boxed{-2} & \boxed{9} \end{array}$$

$$\begin{array}{lll} \textcircled{58} -3(-4+x)^2 - 8, x=3 & \textcircled{59} y^2 + 1, y=3 & \textcircled{60} 4z^2, z=3 \\ -3(-4+3)^2 - 8 & y^2 + 1 & 4(3)^2 \\ -3(-1)^2 - 8 & 3^2 + 1 & 4(9) \\ -3(1) - 8 & 9 + 1 & \boxed{36} \\ -3 - 8 & \boxed{10} & \end{array}$$

$$\begin{array}{lll} \textcircled{61} 3 + x^2, x=1 & \textcircled{62} -5y^2, y=1 & \textcircled{63} z^2, z=-6 \\ 3 + 1^2 & -5(1)^2 & (-6)^2 \\ 3 + 1 & -5(1) & \boxed{36} \\ \boxed{4} & \boxed{-5} & \end{array}$$

$$\begin{array}{llll} \textcircled{64} x^2, x=-3 & \textcircled{65} y^2, y=-1 & \textcircled{66} z^2, z=-8 & \textcircled{67} x^2, x=-4 \\ (-3)^2 & (-1)^2 & (-8)^2 & (-4)^2 \\ \boxed{9} & \boxed{1} & \boxed{64} & \boxed{16} \end{array}$$

$$\begin{array}{llll} \textcircled{68} -y^2, y=3 & \textcircled{69} -z^2, z=4 & \textcircled{70} -x^2, x=2 & \textcircled{71} -y^2, y=1 \\ -3^2 & -4^2 & -2^2 & -1^2 \\ \boxed{-9} & \boxed{-16} & \boxed{-4} & \boxed{-1} \end{array}$$

$$72) -x^2, x = -9 \quad 73) -y^2, y = -5 \quad 74) -z^2, z = -1 \quad 75) -x^2, x = -2$$

$$-(-9)^2 \quad -(-5)^2 \quad -(-1)^2 \quad -(-2)^2$$

$$-1(-9)^2 \quad -25 \quad -1 \quad -4$$

$$-1(81) = -81$$

$$76) -(-(-2)) \quad 77) -(-(-(-3))) \quad 78) 4y^2, y = -4$$

$$-1(-1(-2)) \quad -1(-1(-1(-3))) \quad 4(-4)^2$$

$$-1(2) \quad -1(-1(3)) \quad 4(16)$$

$$-2 \quad -1(-3) = 3 \quad 64$$

$$79) -5z^2, z = -2 \quad 80) 4x^2, x = -1 \quad 81) -3y^2, y = -3 \quad 82) -2z^2, z = 3$$

$$-5(-2)^2 \quad 4(-1)^2 \quad -3(-3)^2 \quad -2(3)^2$$

$$-5(4) \quad 4(1) \quad -3(9) \quad -2(9)$$

$$-20 \quad 4 \quad -27 \quad -18$$

$$83) -6x^2, x = 2 \quad 84) -2y^3, y = -1 \quad 85) -4z^3, z = -2 \quad 86) 2x^3, x = -3$$

$$-6(2)^2 \quad -2(-1)^3 \quad -4(-2)^3 \quad 2(-3)^3$$

$$-6(4) \quad -2(-1) \quad -4(-8) \quad 2(-27)$$

$$-24 \quad 2 \quad 32 \quad -54$$